



CHEMISTRY 3

ACADEMIC

Grade 11/12

Curriculum Guide

CHEMISTRY 3

Prerequisites: Chemistry 1, Chemistry 2

Course Description:

This course builds on the foundational concepts and skills developed in Chemistry 1 and Chemistry 2 by enabling learners to examine acid-base systems and chemical equilibria in aqueous solutions. The learners explore acid-base behavior using Arrhenius, Brønsted-Lowry, and Lewis' theories. They analyze the concentrations of acids and bases, conduct titration experiments, and interpret titration curves to understand how the strengths of reactants influence pH at the equivalence point. The course also examines buffer systems and their role in maintaining stable pH in biological, environmental, and chemical systems. The learners investigate dynamic chemical equilibrium in closed systems and apply equilibrium constants and Le Chatelier's principle to predict reaction extent and direction.

The course also explores separation techniques. The learners use colorimetry to determine concentrations of solutions and interpret experimental data. They explain oxidation-reduction processes, conduct simple electrolysis experiments, and analyze applications of redox reactions in electrochemical cells, batteries, corrosion prevention, and sustainable energy technologies.

The learners analyze experimental and graphical data, apply mathematical models, and communicate scientific explanations. The course promotes systems thinking in analyzing the applications of chemistry in environmental science, health sciences, and industrial processes, particularly those related to the Sustainable Development Goals and Green Chemistry.

Academic: Elective

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING STANDARDS <i>The learners...</i>
Acid-Base and Chemical Equilibrium		
1. Strength and Concentration of Acids and Bases	1. acids and bases found in everyday life and environmental systems play important roles in maintaining chemical balance in natural environments and human activities; 2. different acid-base theories explain the behavior of acids and bases; 3. the pH of solutions describes the concentration of hydrogen ions, and that the strength and concentration of acids and bases determine their behavior in chemical reactions;	1. differentiate Arrhenius, Brønsted-Lowry, and Lewis' acid-base theories;
		2. calculate the pH of acids and bases commonly found at home and in the environment;
		3. conduct pH measurements to identify differences in the strengths of acids and bases;
		4. describe the dissociation of acids and bases in water, conjugate acid-base pairs in solution, and the amphoteric nature of certain substances;
		5. explain the difference between strong and weak acids and bases and distinguish these from concentrated and dilute solutions;
2. Acid-Base Titrations	4. titration curves provide graphical representations of acid-base reactions;	6. conduct simple titration experiments and perform titrimetric calculations;
3. Buffer Systems	5. buffer systems maintain relatively constant pH in natural, biological, and chemical systems; and	7. explain the role of buffer systems in maintaining stability in natural environments and biological systems;
4. Chemical Equilibrium	6. chemical systems reach dynamic equilibrium in closed systems and that the extent and direction of reactions can be predicted using equilibrium	8. differentiate static and dynamic equilibrium and relate equilibrium to closed chemical systems;
		9. explain the reversibility of chemical reactions using real-life examples such as the dissolution and release of carbon dioxide in water, the formation and

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	constants and Le Chatelier’s principle.	decomposition of calcium carbonate in limestone, and the Haber-Bosch process for ammonia synthesis;
		10.use equilibrium constant expressions (Kc) to calculate the concentrations of reactants and products at chemical equilibrium and determine the extent and direction of a chemical reaction;
		11.conduct investigations on the effects of temperature, concentration, volume or pressure on a chemical system at equilibrium; and
		12.analyze how reaction rates and chemical equilibrium are applied in environmental processes and industrial practices to optimize processes, reduce waste, and design sustainable solutions using Green Chemistry and systems thinking.
Performance Standards <i>The learners</i> can design and conduct investigations to analyze the properties and behavior of acids, bases, and buffer systems. They interpret pH values and titration curves using mathematical and graphical models and communicate scientific explanations using chemical equations and experimental evidence. The learners can also explain chemical equilibrium and Le Chatelier’s principle, conduct investigations to observe changes in equilibrium systems, and apply these concepts to explain industrial and environmental processes using Green Chemistry principles.		
Suggested Performance Tasks - Conduct an acid-base titration investigation to determine the concentration and strength of an unknown acid or base commonly found in household or environmental samples. Explain the results using acid-base theories and neutralization concepts. - Conduct an investigation on chemical equilibrium by changing temperature, concentration, or pressure in a reversible reaction system using Le Chatelier’s principle.		

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Instrumental Analysis and Electrochemistry		
5. Separation of Mixtures	7. mixtures can be separated based on differences in physical properties using appropriate techniques;	13. differentiate common separation techniques, such as filtration, recrystallization, distillation, and chromatography based on their principles;
		14. conduct a laboratory activity using a separation technique and explain the results based on physical properties;
6. Instrumental Analysis (Colorimetry)	8. instrumental analytical techniques use measurable signals such as light absorption to determine the concentration of substances;	15. explain the principles of colorimetry based on light absorption by substances;
		16. describe the relationship between absorbance and concentration using calibration curves;
		17. conduct a colorimetry-based investigation to determine the concentration of an unknown solution;
7. Redox Reactions	9. oxidation-reduction reactions involve electron transfer and changes in oxidation numbers;	18. use the concepts of oxidation and reduction in terms of electron transfer and oxidation numbers to analyze redox reactions;
		19. write half-reactions and balance overall equations to represent redox reactions;
8. Galvanic Cells	10. electrochemical cells convert chemical energy into electrical energy through redox processes;	20. create diagrams of galvanic cells to identify the anode, cathode, and direction of electron flow;
9. Electrochemical Potentials	11. standard electrode potentials and cell potentials can be used to predict the direction and spontaneity of redox reactions;	21. use table of standard reduction potentials to identify and compare the reduction potentials of galvanic half-cells and the reactivity of metals in aqueous solutions;
		22. use cell potential values to predict the spontaneity of redox reactions;

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10. Electrolysis	12. electrolysis drives nonspontaneous redox reactions using electrical energy; and	23. conduct experiments on electrolysis processes; and
11. Applications of Electrochemistry	13. electrochemical principles are applied in technologies such as batteries, corrosion control, and sustainable energy systems.	24. explain electrochemistry applications in batteries, sustainable energy technologies, and corrosion control.
Performance Standards <i>The learners can explain different separation techniques, interpret colorimetric data to determine concentrations, and conduct investigations using appropriate techniques. The learners can analyze redox reactions and use standard reduction potentials and cell potential values to predict reactivity and spontaneity. They can also explain electrolysis processes and apply electrochemical principles to real-world systems such as batteries, corrosion control, and sustainable energy technologies.</i>		
Suggested Performance Tasks • Perform laboratory activity using either a separation technique (e.g., filtration or chromatography). • Conduct a colorimetry investigation to determine the concentration of an unknown solution using a calibration curve and interpret the results. • Construct and analyze a simple galvanic cell using two different metal electrodes and electrolyte solutions and explain the electron flow and spontaneity of the reaction using redox concepts. Investigate electrochemical applications by analyzing a real-world system (e.g., battery technology, corrosion prevention, or metal electroplating) and explain how electrochemical principles are used to improve energy efficiency, prevent corrosion, or support sustainable technologies.		